

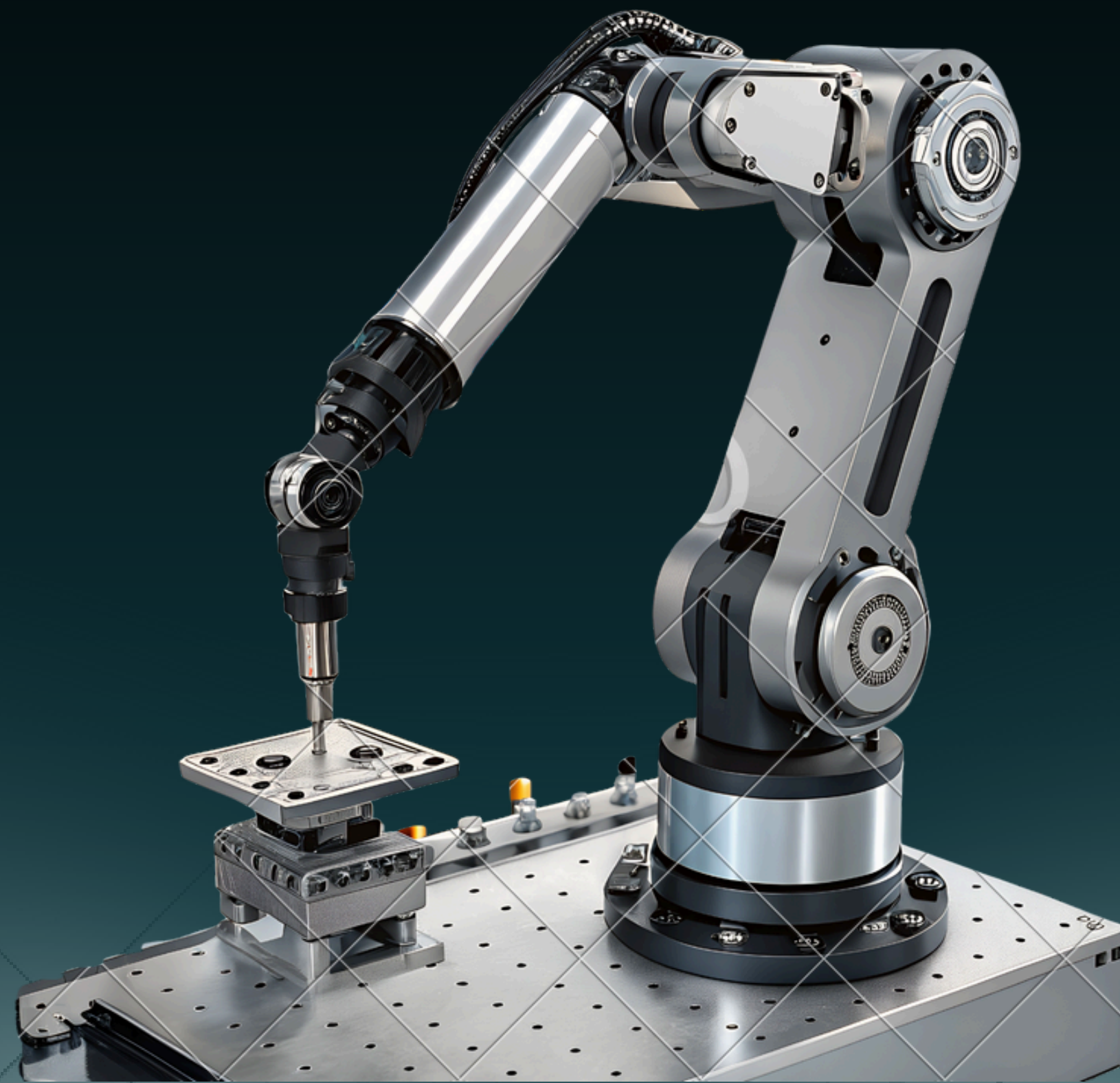
2026



PNT
ACADEMY

PRESENTS:

RESEARCH- ENABLED INDUSTRIAL AUTOMATION LAB



PROBLEM STATEMENT

Here are the problems in summary:



- China's "Made in China 2025" initiative heavily promotes industrial automation.

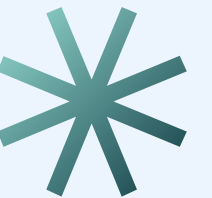


- Result: China produces 150,000+ robotics graduates annually, who seamlessly integrate into the workforce.



- Colleges partner with robotics manufacturers to offer students live project experience and internships.

- A study by NASSCOM reveals only 10% of Indian engineering graduates are employable in robotics-related fields.
- Theoretical-heavy curriculum fail to provide hands-on experience and industry-ready skills.
- India highly lags behind countries like China and Japan in the adoption of AMR, AGVs and robotic arms.



THE OPPORTUNITY



Permanent Industrial
Robotics Lab Setup



Curriculum-
Integrated Learning
Model



Faculty Enablement
& Ownership

Student Skill
Development &
Industry Exposure



Innovation, Research
& IIC Enablement

PNT Academy proposes the establishment of a permanently deployed, industry-aligned Robotics & Autonomous Systems Lab within the institution, designed to bridge curriculum gaps, enhance employability, and support innovation, research, and entrepreneurship 



OUR SOLUTION

That's where we come in—with an industry-aligned Robotics & Autonomous Systems Lab within the institution, designed to bridge curriculum gaps, enhance employability, and support innovation, research, and entrepreneurship.



Robotic Arm



Industrial Robotic Arm



Industrial Graded AGV



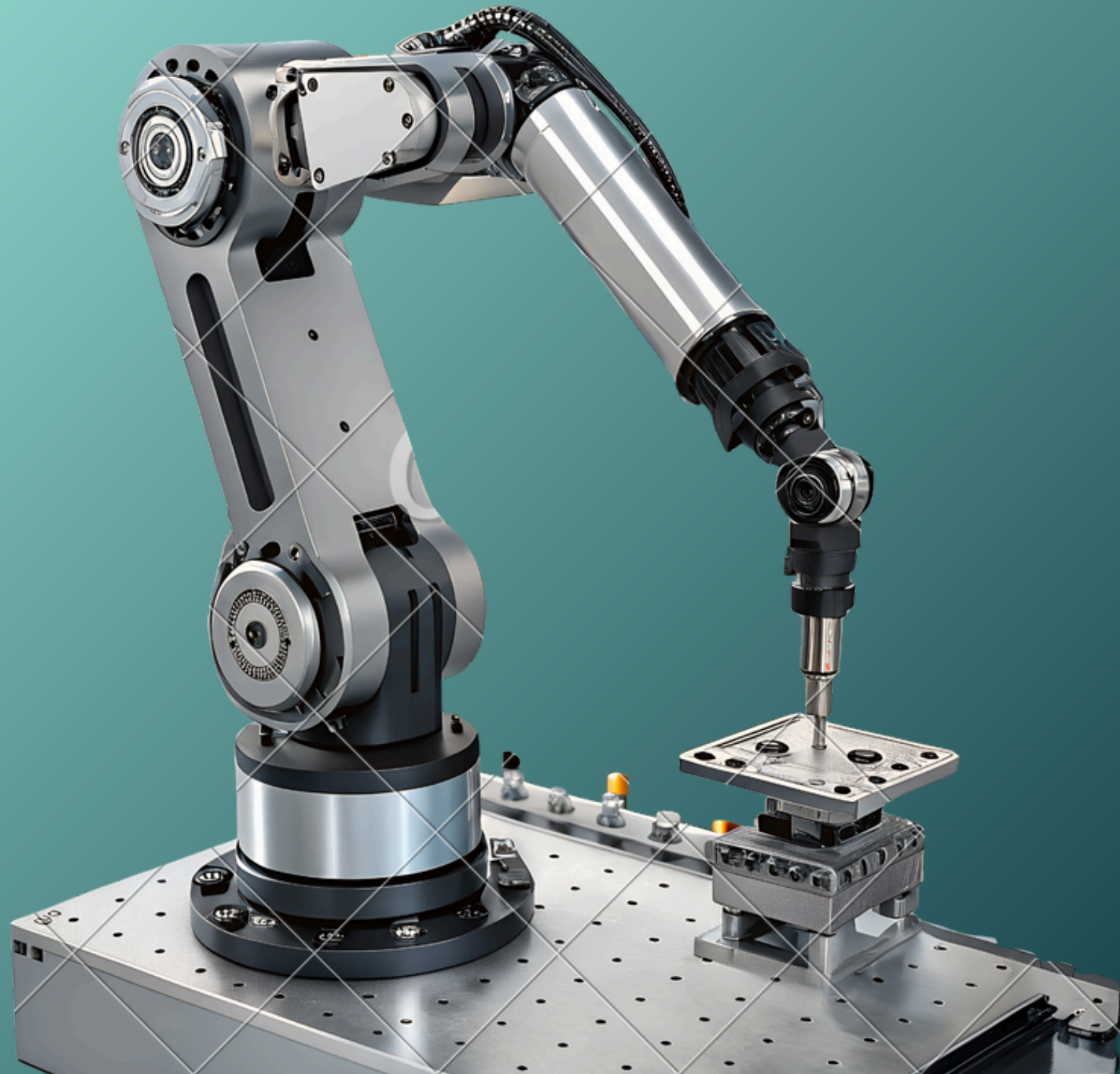
Robotic Hand



Drone



Autonomous Mobile Robot (AMR)



HOW IT WORKS

- Deploy any of the package permanently at your institution.
- Free hands-on 30 hours training for students and faculty
- Industrial exposure to real-world applications in logistics, smart warehouses, and supply chain automation.
- Internship to students on Indian Navy, Indian Army projects with PNT Robotics
- LOR to selected students
- Entry in the Robotics Competition

VALUE PROPOSITION



Modular Design:

- Add or replace microcontrollers / SBCs
- Integrate additional sensors, drivers, and communication modules
- Increase degrees of freedom in robotic arms
- Redesign power distribution, control logic, and feedback loops

3D Printed & Customizable Structure:

- Some part of the robotic arm , AGV OR AMR will be 3D printed, enabling students to redesign and modify its structure.
- CAD files and tutorials will be provided so students can experiment with different designs and print their own parts.

Open-Source Software & Coding Flexibility:

- Robotic Arm + AMR integration
- AGV + Robotic Hand coordination
- Custom hybrid robotic systems for research projects

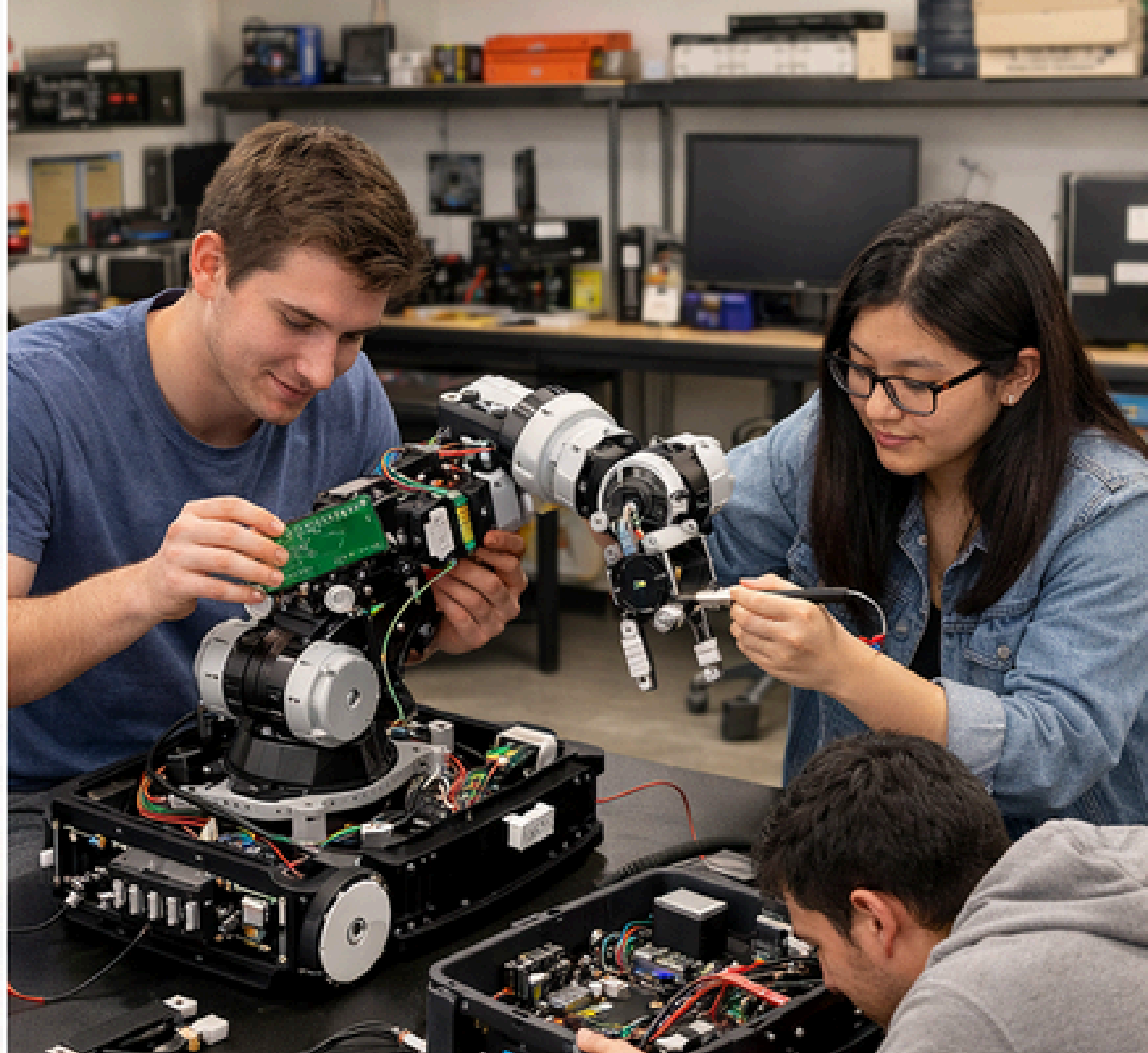
Research & IIC Enablement

- Encourages:
- Reverse engineering
- System-level thinking
- Innovation beyond predefined experiment

Traditional Setup



Customizable Setup



EDUCATIONAL ROBOTIC ARM

4 DOF robotic arm, customizable components, 3D printed replaceable parts.

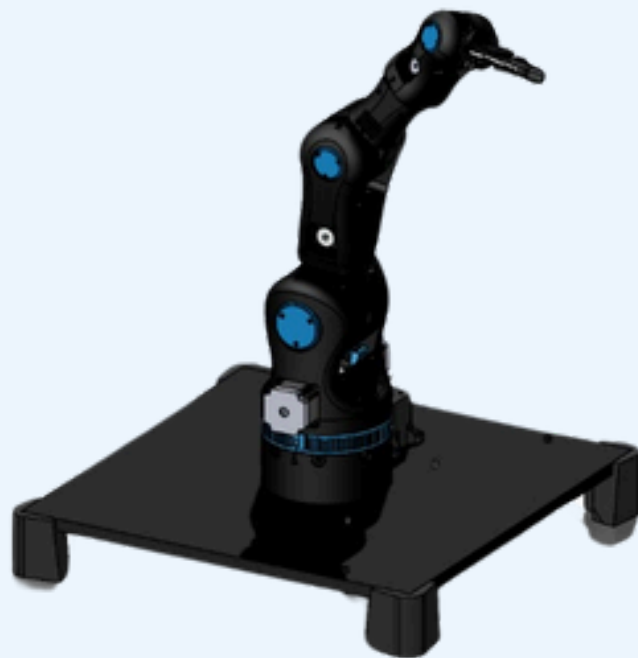


- * Demonstration of pick and place Automation
- * Demonstration of gesture-controlled robotic arm
- * Sensor Feedback Integration with robot Arm
- * Path Planning and Motion Control
- * Object Detection using OpenCV
- * Object Sorting by Colour & by Size
- * Programming for Industrial Applications
- * Integration with IoT for Remote Control
- * AI integration with robot Arm

Unlike plug-and-play labs, our robots are fully open—enabling circuit-level customization, DOF expansion, and multi-robot integration

INDUSTRIAL ROBOTIC ARM

4–6 Axis articulated robotic arm,
Payload capacity: 1 – 2 kg (education &
lab grade), Repeatability accuracy up to
 ± 0.1 mm

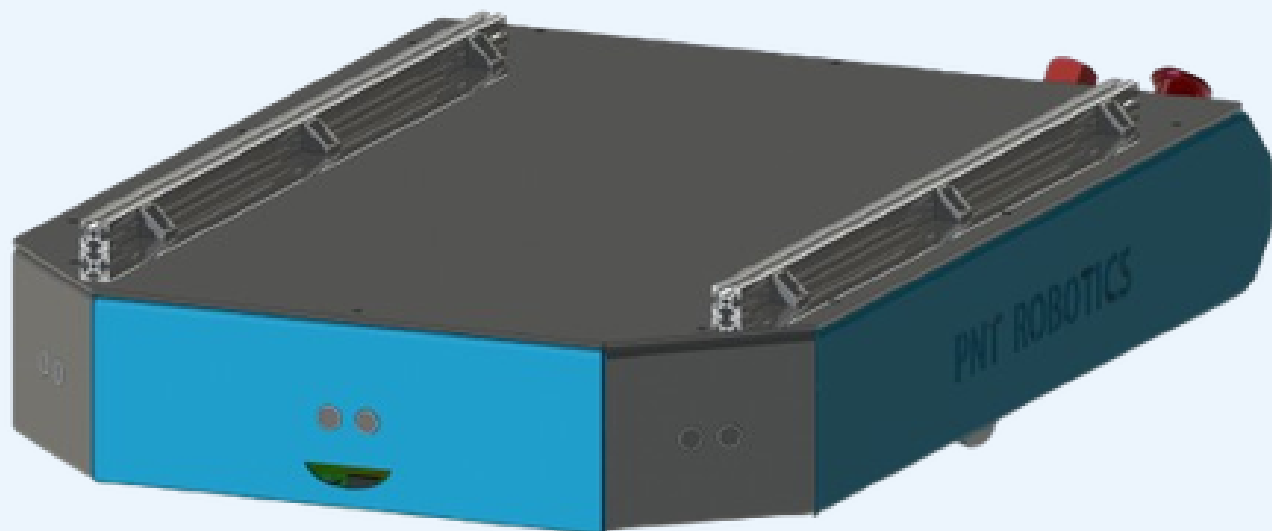


- * Forward & inverse kinematics programming
- * Pick-and-place task execution
- * Tool path planning for repetitive tasks
- * Payload testing and accuracy calibration

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INDUSTRIAL AGV

Industrial-grade AGV, custom-design capabilities, integration with smart factories.



- * Demonstration of line following AGV
- * Demonstration of obstacle avoidance in AGV
- * Demonstration of RFID-based navigation system in AGV
- * Demonstration of automated goods transportation in AGV
- * Speed and acceleration control in AGV
- * Wireless Control of AGV
- * IoT-based Control of AGV
- * Path Planning based on custom designed algorithms
- * Battery and Power Optimization in AGV
- * Integration of external sensors in AGV

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ROBOTIC HAND

Multi-finger robotic hand, customizable parts, gesture-controlled interface.



- * Demonstration of finger movement coordination
- * Demonstration of feedback mechanism
- * EMG-based hand control
- * Gesture-based hand control
- * Prosthetic Hand Demonstration
- * Haptic Feedback System
- * AI-Based control of hand
- * AI based gesture control of hand
- * Integration with Wearable Tech
- * Remote-controlled hand movement

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INDUSTRIAL DRONE

Programmable drones with custom control options.



- * API usage and calibration
- * Sensor data collection and usage in drone
- * Drone take off and landing automation
- * Obstacle avoidance using object detection
- * Warehouse automation using barcodes/QR codes
- * Aerial Photography and Surveillance
- * Face tracking with drone flight
- * AI-Based Object Detection from Aerial View
- * AI Surveillance drone with Intruder detection
- * Development of PC application to control drone

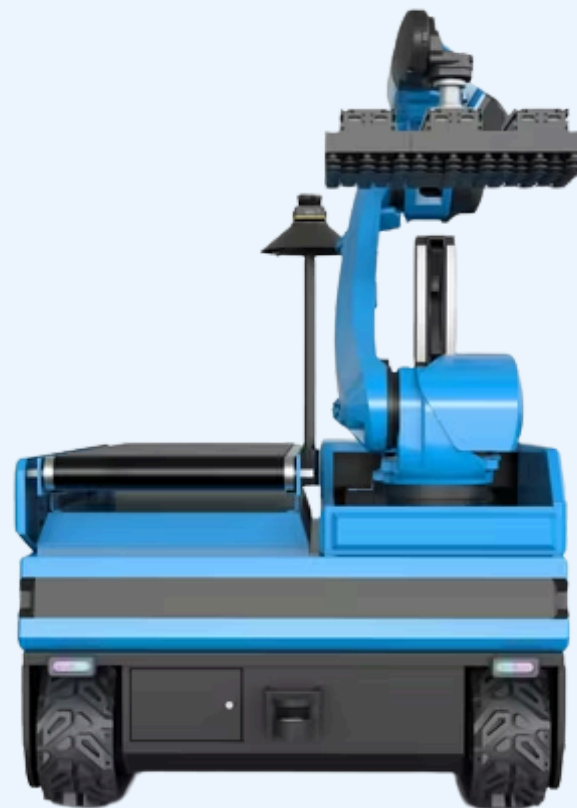
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INDUSTRIAL AMR

Differential / Mecanum wheel drive system, On-board microcontroller / SBC (Raspberry Pi / equivalent), Li-ion battery with 2–3 hours operational time

Sensors: Ultrasonic, IR, IMU, optional LiDAR

Supports ROS-based & block-based programming



- * Hardware interfacing and integration of sensors and actuators with the Robot Operating System (ROS).
- * Utilizing odometry sensor data and implementing teleoperation control for Autonomous Mobile Robots (AMRs).
- * Mapping a defined indoor environment using LiDAR sensors.
- * Implementing autonomous navigation with parameter tuning to optimize navigation performance.
- * Addressing real-world challenges in logistics and defense deployment scenarios.

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COLLEGE BENIFITS

- Supports NEP 2020 experiential & skill-based learning
- Innovation challenges
- Prototype development
- Hackathons
- Student startups

- FDPs for faculty (paid / sponsored)
- Certification programs for external students
- Industry workshops & bootcamps
- Sponsored final-year projects

- Train-the-trainer certification
- Faculty access to:
 - Hardware schematics
 - Software repositories
 - Experiment manuals
 - Joint mentoring model (PNT + faculty)

- Capstone & minor project themes
- IEEE / Scopus paper guidance
- Prototype-to-patent pathway
- Support for innovation grants & competitions

BRANCH ALIGNMENT & CURRICULUM

Mechanical, Mechatronics, Electrical,
E&TC, AI-ML, Robotics



* Robotics & Automation

* Control Systems

* Mechatronics

* Embedded Systems

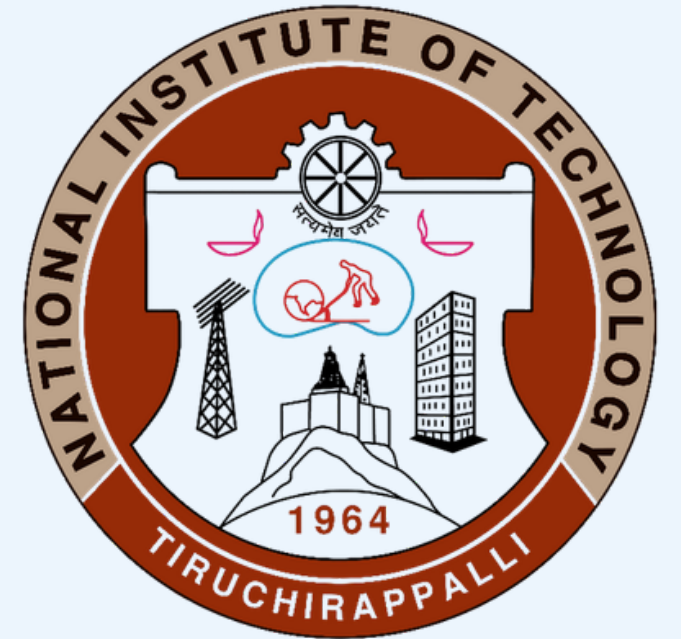
* ROS & Autonomous systems

OUR CLIENTS



SOMAIYA
VIDYAVIHAR UNIVERSITY

VIT | Vidyalkar
Institute of
Technology



COMPANIES HIRING FOR AGV & ROBOTIC ARM

1.ABB Mobile Robotics	11.PNT Robotics
2.Fanuc India	12.Mitsubishi Electric India
3.Tata Elxsi	13.Wipro Engineering
4.Bosch India	14.Cognizant Technology Solutions
5.Larsen & Toubro (L&T)	15.Indian Oil Corporation (IOCL)
6.Siemens India	16.Bluebotics India
7.GrayOrange Robotics	17.Industrial Robotics
8.KUKA Robotics India	18.Heavy-Duty AGVs
9.Honeywell Automation	19.Collaborative Robotics
10.Schneider Electric India	20.General-Purpose Robotics

OUR TOP INDUSTRY ASSOCIATIONS

Indian Army



TATA Power

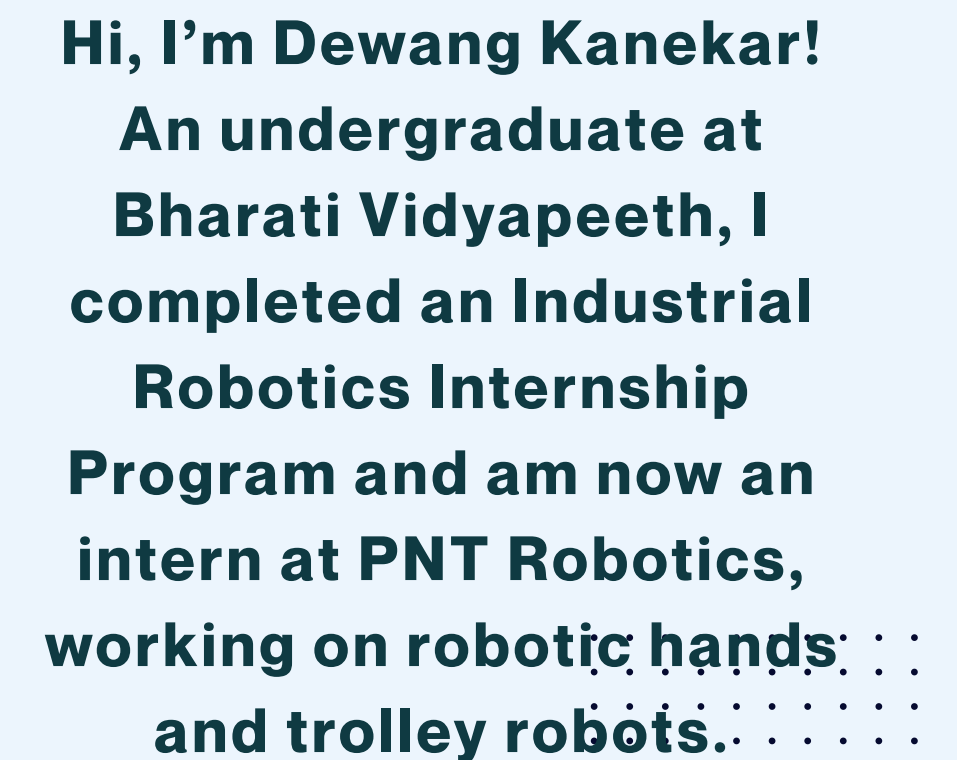
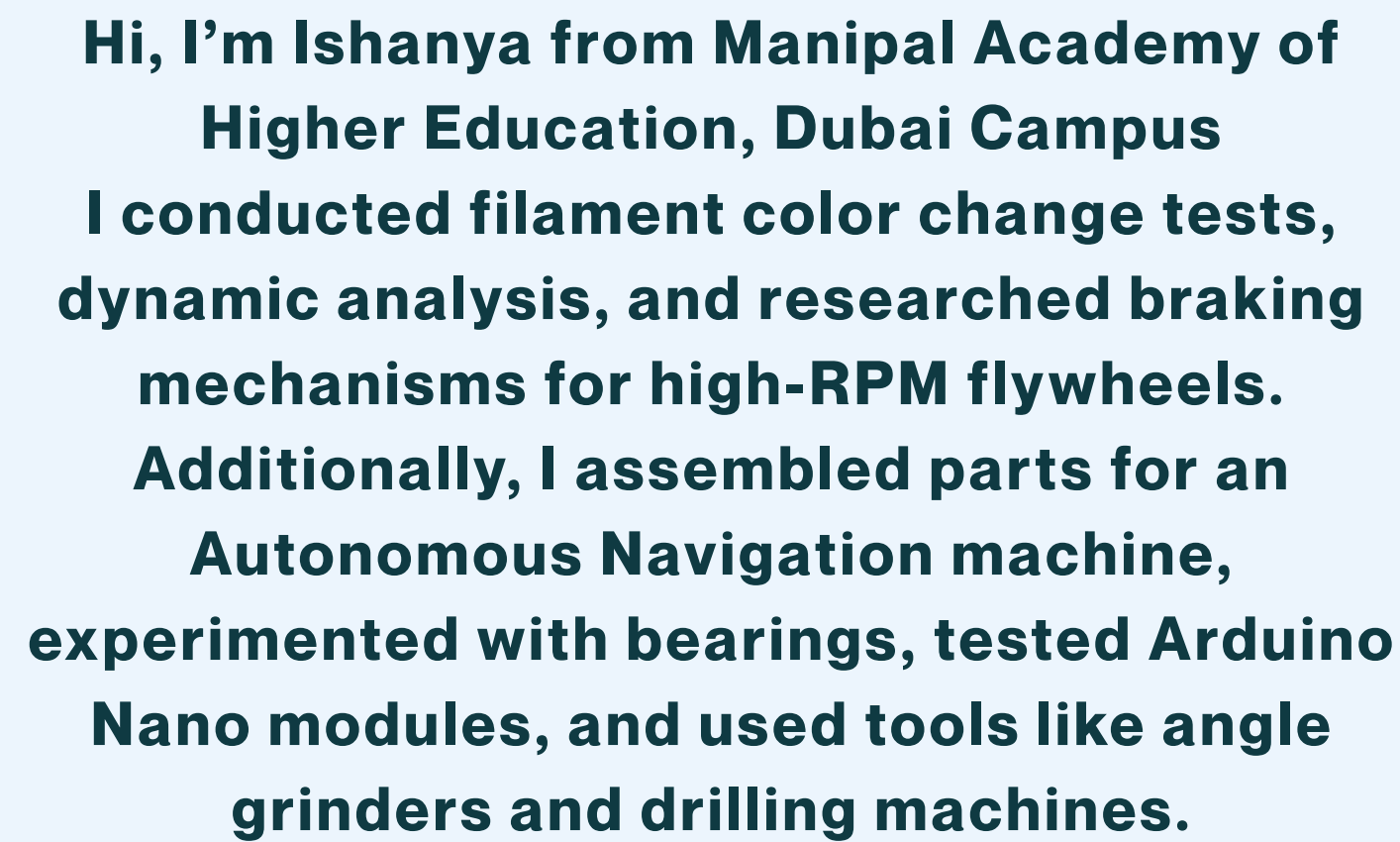


Wockhardt



Unilever UK

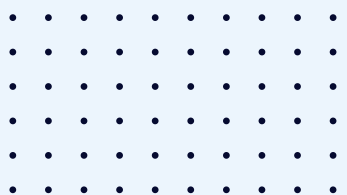




Conclusion

PNT Academy invites your institution to be a part of this transformative journey in automation education. Together, let us shape the future of education and innovation.

Join hands with PNT Academy to revolutionize your institution's approach to learning and empower the next generation of automation leaders.





GET IN TOUCH

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